GAS TURBINES

A prime mover, produces mechanical power from heat energy .

Gas turbine is classified into two types.

i) Open cycle GT

ii) Closed cycle GT

Open cycle GT is basically an "Internal Combustion Engine", since combustion products directly expand in the turbine.

The thermodynamic cycle used for analysis of gas turbine is named after George Brayton and called "Brayton cycle".

Applications:

Aircrafts (high power to weight ratio)

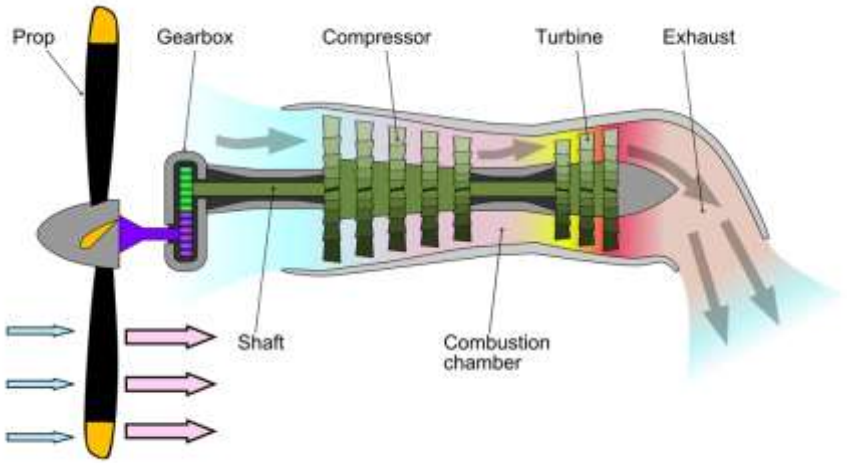
Oil and gas industry

Power generation

AIRCRAFT GAS TURBINES

"Turboprop engine" is a gas turbine engine which drives an aircraft propeller using a reduction gear.

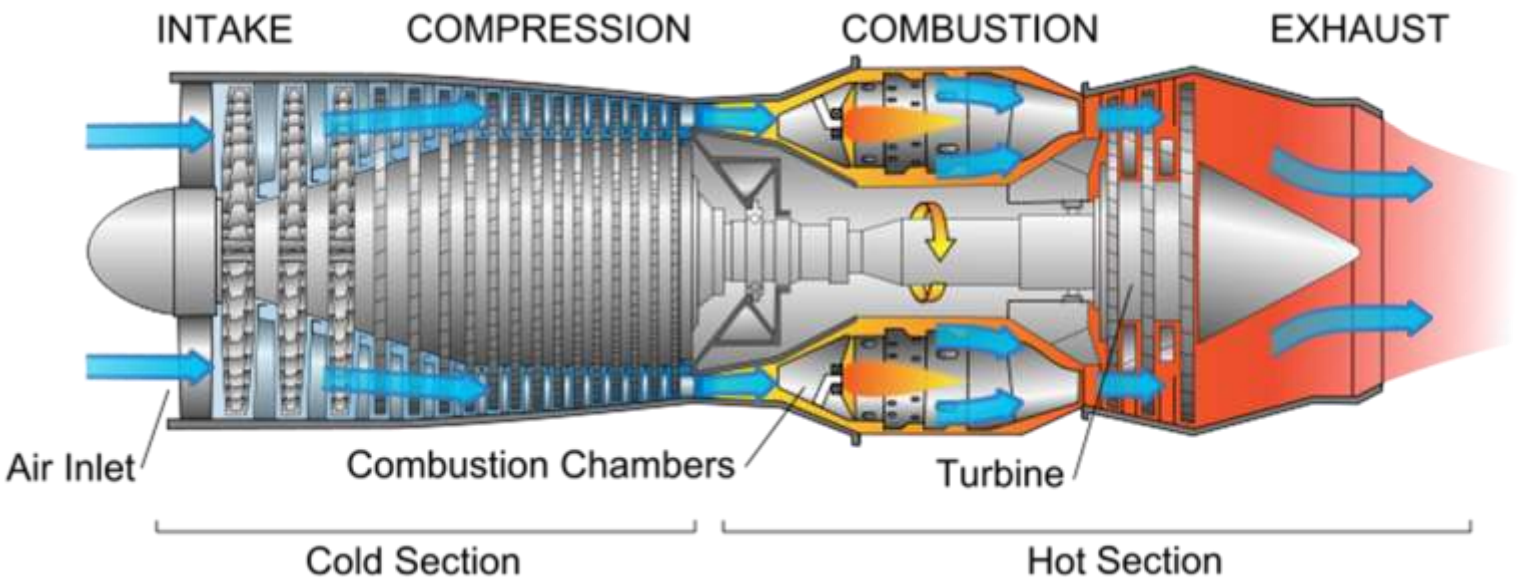
The gas turbine output is used to drive the propeller.





AIRCRAFT GAS TURBINES

"Turbojet engine" is a gas turbine engine with a propelling nozzle at the end where the gas is accelerated to high speed to provide the required thrust.



GAS TURBINES MANUFACTURERS

Mitsubishi (Japan)

Hitachi (Japan)

Pratt and Whitney (USA)

General Electric (USA)

Rolls Royce (UK)

Siemens (Germany)

Alstom (France)

OPEN CYCLE GAS TURBINE

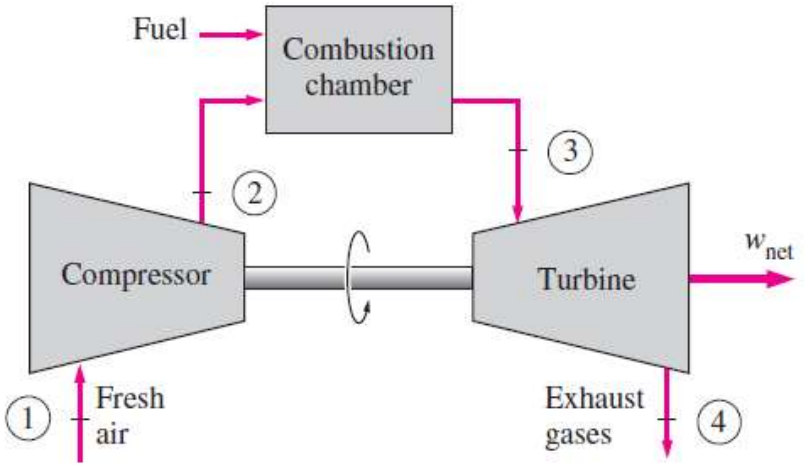
The major components are,

- Compressor
- Combustion chamber (CC)
- Turbine.

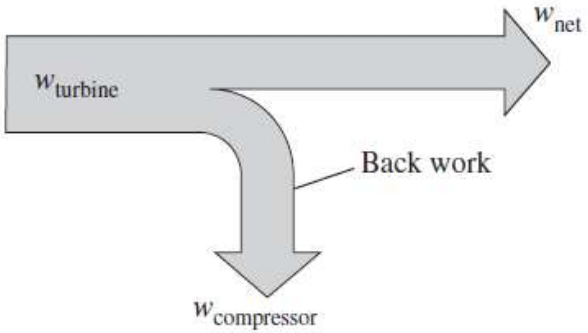
Atmospheric air enters the compressor and is compressed (p&T ↑).

In the CC, the fuel (gas, liquid or pulverised solid) is burnt and high press. & high temp. gas is produced.

The gas expands in the turbine and produces mechanical work.



BACK WORK RATIO



The compressor requires mechanical power for its drive.

The ratio of the compressor work to the turbine work, called "Back work ratio", is very high in gas turbines.

Around 50% of the power produced by the turbine is consumed by the compressor.

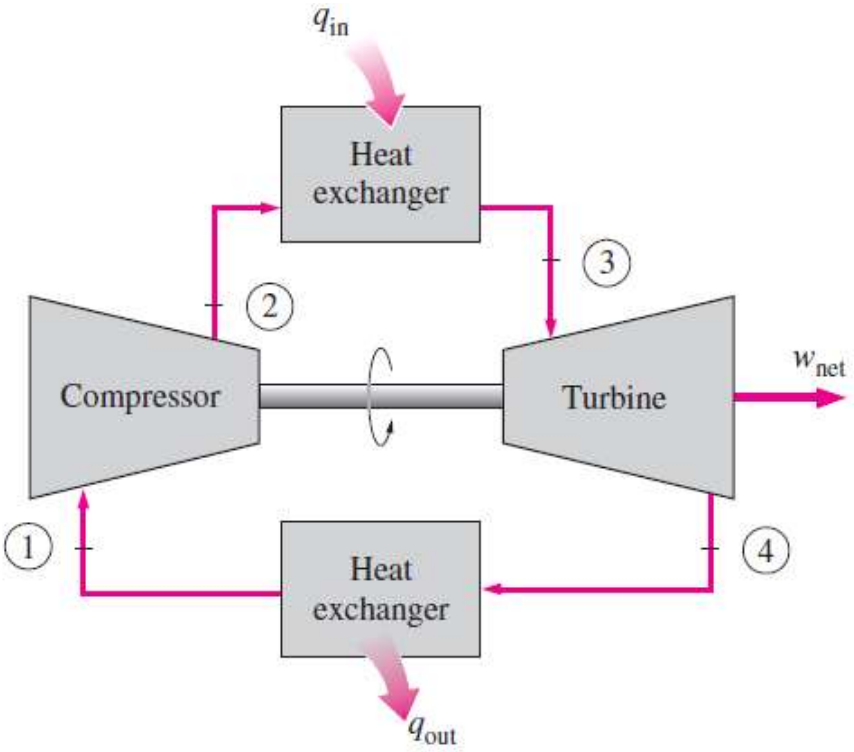
Hence, a starting system is required.

CLOSED CYCLE GAS TURBINE

The working fluid is not discharged into the atmosphere after the turbine.

Instead, it flows through a heat exchanger, gives away some heat and enters the compressor as a cooled gas.

By cooling the gas, the volume is reduced and thus the compressor work is minimized.



CLOSED CYCLE GAS TURBINE

After the compressor the gas passes through another heat exchanger (High temperature) where its temperature is increased.

This hot gas then enters the turbine and the cycle of operation continues.

The working medium is the same gas (need not be air) and it remains uncontaminated by the products of combustion.

Brayon Cycle (P-v and T-s diagrams)

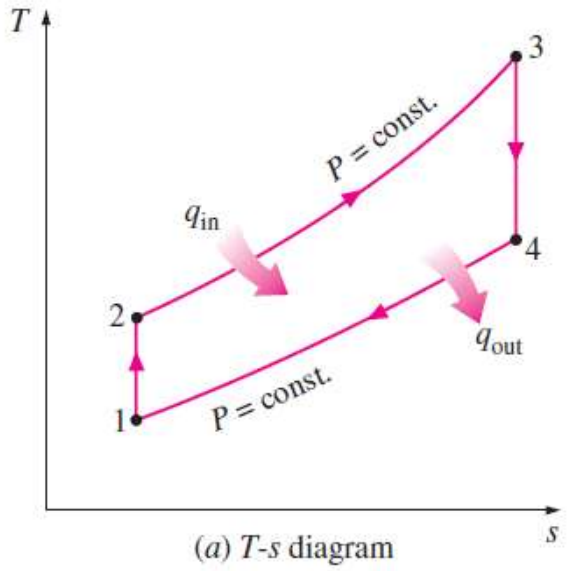
The ideal cycle is made up of four internally reversible processes:

1-2 : Isentropic compression (in a compressor)

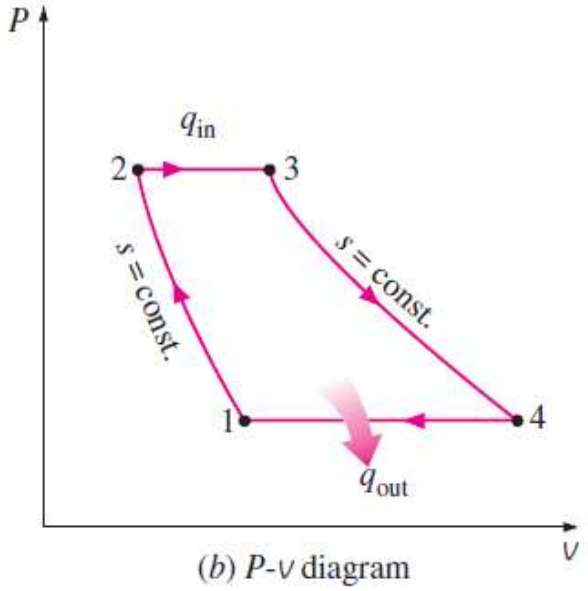
2-3 : Constant-pressure heat addition

3-4 : Isentropic expansion (in a turbine)

4-1 : Constant-pressure heat rejection



(a) T-s diagram



(b) P-v diagram